Do Now · retrieval

Lesson 112 · Choosing Applications and Utility Software

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What role does a device driver perform?

Q2

A linked-list node has data and what other essential field?

Q3

In a BST, where is a value less than the current node placed?

Q4

Why can an array provide direct indexed access?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Clarify needs — Identify users, scale, accessibility, platform and constraints.

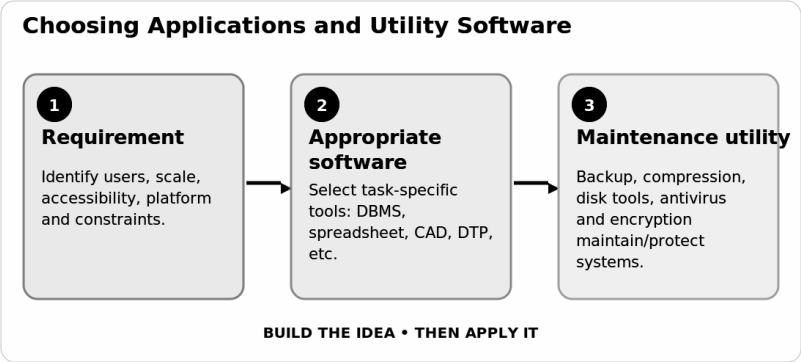
Core idea 2

Match applications — Select task-specific tools: DBMS, spreadsheet, CAD, DTP, etc.

Core idea 3

Understand utilities — Backup, compression, disk tools, antivirus and encryption maintain/protect systems.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

application software — Programs that meet a user task.

utility — Software for maintenance, protection or optimisation.

DBMS — Database management system.

backup — Recoverable copy to protect against data loss.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

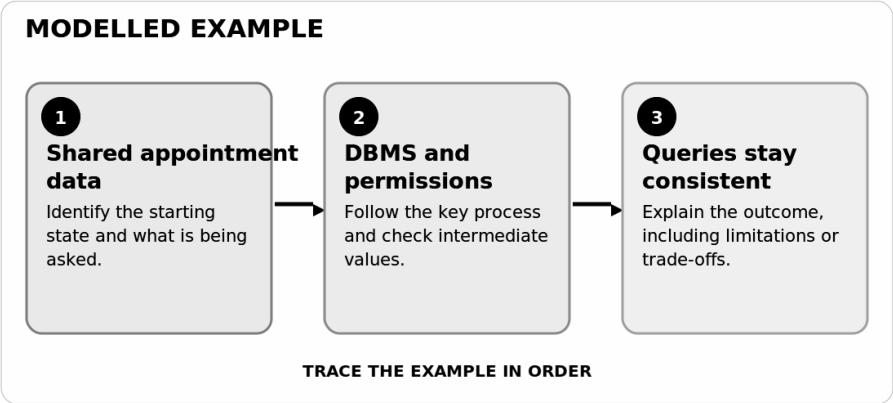
Problem

A clinic manages thousands of appointments accessed concurrently. Spreadsheet or database application?

Follow the worked method

1. Model answer — A DBMS is better: structured records, querying, concurrency, constraints and access rights. A spreadsheet is useful for small-scale analysis but is not ideal as a multi-user transactional store. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Justify an application for editing a school magazine with text and complex page layout. [6]

Question 2

Differentiate application software from utility software using two examples. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A small charity needs inventory records, backups and accessible user interfaces. Recommend software and justify limitations. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A backup utility and a word processor are both examples of utility software.

Use key vocabulary from memory

1. Explain application software in your own words.

2. Explain utility in your own words.

3. Explain DBMS in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.